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A translanguaging perspective on students' use of generative artificial intelligence in L2 writing

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ABSTRACT

Generative artificial intelligence (GAI) has been acknowledged as a valuable resource in second language (L2) writing but often researched on as a source of monolingual support in the target language in quasi-experimental settings. Students' use of GAI in naturalistic L2 writing contexts with reference to their multilingual and multimodal repertoires, however, remains underresearched. To address the gap, this study draws on translanguaging as the theoretical lens and explores the ways in which Chinese multilingual undergraduates used GAI as a linguistic support and the underlying reasons shaping their use. Three data sources were employed: screen recordings, reflective journals, and individual semi-structured interviews. The data were analyzed using thematic analysis. The findings revealed that the students used GAI in four main patterns: creation for generating writing prompts, outlines, and content; translation for comprehension and expression; evaluation for improvement and decision-making; and revision for language and content. Three distinct stances have emerged as key factors shaping students' GAI use: participation stances, language stances, and tool stances. The findings suggest that GAI-assisted L2 writing can be theorized as a distributed translanguaging practice between students, GAI and other tools that creates a space fostering students' criticality and creativity. Educators are thereby recommended to respect students' individualized translanguaging use of GAI and cultivate their translanguaging literacies, enabling students to use GAI more effectively based on their L2 writing needs.

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Generative artificial intelligence; L2 writing; translanguaging; students

1. Introduction

For many students, effectively producing second language (L2) texts that accurately express first language (L1) meaning remains a significant challenge in L2 writing. As a result, they commonly turn to online dictionaries or neural machine translation tools for translanguaging support (Jiang et al., 2024; Vinall et al., 2024). However, these translanguaging supports are increasingly being replaced by the growing prevalence of generative artificial intelligence (GAI) (Yang & Lin, 2025), which offers superior

translation quality (Lee, 2024) and assists students in generating higher-quality L2 writing (Kim & Chon, 2025).

Students' interactions with GAI tools reveal a complex form of shared agency that involves facilitation and inhibition (Godwin-Jones, 2024). GAI functions both as a versatile assistant and a barrier in L2 writing. While GAI enhances writing productivity, it also risks displacing the cognitive labor essential to linguistic development, which may lead to the emergence of passive and uncritical writers devoid of agency (Shi et al., 2025; Zou & Huang, 2023). Like other translation tools, its automated features can diminish learners' motivation to use their L2 actively, thereby weakening their linguistic competence (Murtisari et al., 2024). Some students treat GAI-generated content as model texts to strengthen their L2 proficiency (Liu et al., 2024). However, this practice may also reinforce linguistic normalization, described as 'a form of cultural imperialism, imposing a standardized language at the expense of diverse linguistic identities' (de Roock, 2024, p. 595). As a result, students' writing may become homogenized and lose its distinctive authorial voice (Zou & Huang, 2023). In response, some students believe that GAI use should be limited (Yan, 2023). Previous critiques have largely conceptualized GAI as a mechanical L1-L2 converter that facilitates code-switching in quasi-experimental settings. What these views often miss is GAI's potential to encourage students to draw on their full linguistic repertoires for solving writing challenges. The effectiveness of GAI in L2 writing depends significantly on students' ability to mobilize their multilingual and multimodal resources in contextually appropriate and agentive ways. This highlights the need for further research that reexamines the relationship between students and GAI through a translanguaging perspective, focusing on how students deploy their linguistic repertoires to negotiate shared agency with GAI. Drawing on this orientation, the present study investigates how students interact with GAI in naturalistic settings, aiming to support more effective use of such tools in L2 writing. It addresses two central research questions: (1) How do students use GAI in naturalistic L2 writing settings? and (2) What factors shape students' use of GAI in L2 writing?

2. Literature review

2.1. *Translanguaging and GAI*

Translanguaging theory in L2 learning emphasizes students' ability to use their full linguistic and semiotic repertoires flexibly and freely in the process of meaning-making (Li, 2018). This process involves transcending the boundaries of named languages and other semiotic limitations. When engaging in translanguaging, students select the features from their communicative repertoires that 'seem to be the most appropriate for the communicative task at hand' (García & Otheguy, 2021, p. 15). As students progress in their L2 learning, they also expand their unitary repertoire by incorporating new elements (García & Otheguy, 2021). Moreover, translanguaging fosters students' creativity and criticality, allowing them to challenge conventions and norms with a critical disposition (Li, 2011). It also offers an 'equitable view of multilingualism' (Li, 2022, p. 10). It positions L2 learners as competent language users with all languages they know playing a role in the L2 learning process.

Situated within cognitive (e.g. Kim & Chon, 2025; Shi et al., 2025) and socio-cultural (e.g. Wang, 2024; M. Zou et al., 2025) paradigms, GAI is recognized for its capacity to assist L2 writers in various ways, such as facilitating the conversion of L1 input into L2 output (Yu & Jiang, 2025; M. Zou et al., 2025). Yet, this body of research treats GAI as a monolingual scaffold for code-switching, emphasizing its assistance with L2 while overlooking its inherently multilingual nature and the multilingual processes students engage in when using it. When struggling to express themselves in L2, students turn to GAI, leveraging their L1 to request translations, interpret system outputs, and generate writing ideas (Yang & Lin, 2025). GAI also supports students' multilingual development by facilitating comparisons between L1 and L2 (Moorhouse et al., 2024). This translingual nature of GAI use challenges deficit-based views of L2 learners, positioning them instead as strategic multilingual users who draw flexibly on their linguistic repertoires in L2 writing. When GAI use is framed primarily around L2 output or monolingual assistance, it risks being perceived as a shortcut that discourages students' critical thinking and as a threat to L2 development (Zou & Huang, 2023). This perspective underestimates students' agency and the complex translanguaging practices involved. A translanguaging lens, by contrast, highlights GAI not as a remedy for linguistic deficiency, but as a multilingual asset students agentively draw upon, supporting students in activating and orchestrating diverse language resources for L2 writing. Despite its conceptual significance, empirical inquiry into GAI use through this lens remains limited (e.g. Yang & Lin, 2025). Accordingly, this study aims to explore how students use GAI as a multilingual resource and to illuminate their perspectives on its role in L2 writing.

2.2. GAI use in L2 writing

Research on GAI use in L2 writing has developed along three primary lines: outcome analysis, perception-based inquiry, and behavioral observation. The outcome-oriented studies have primarily examined whether and how GAI affects students' L2 writing performance (Kim & Chon, 2025; Shi et al., 2025). Experimental findings from comparisons among GAI-assisted, machine translation-assisted, and unaided writing conditions show GAI-assisted writing produced greater lexical diversity, syntactic complexity, and overall accuracy than writing without AI assistance (Kim & Chon, 2025). However, it also resulted in lower verb cohesion than the other two conditions (Kim & Chon, 2025). Other work has examined the relative writing capacities of students and GAI (Zhou et al., 2023). Comparative analyses between student-authored texts and GAI-generated drafts found that GAI outperformed students in narrativity, word concreteness, referential cohesion, but fell short in deep cohesion (Zhou et al., 2023). Nevertheless, the emphasis in these studies remains on monolingual L2 performance analysis, with limited attention given to the multilingual dimensions of students' translanguaging practices during writing.

A considerable body of research has focused on students' perceptions of GAI use in L2 writing. Some of these investigations have examined the perceived effects of GAI on writing (Kim et al., 2025; Shi et al., 2025; Yang & Lin, 2025; Zou & Huang, 2023). Through interviews, researchers have uncovered advantages related to the

writing process, writing performance, and affective engagement, such as increased enjoyment (Kim et al., 2025). Other research analyzing students' reflective papers demonstrates that GAI offers practical benefits including enhanced brainstorming but may also present challenges such as obstructing L2 learning (Zou & Huang, 2023). Researchers have also examined students' suggestions and preferences for how GAI ought to be integrated into L2 writing (Kim et al., 2025; Zou & Huang, 2023). Students have expressed their expectations for GAI to function as a multi-tasking assistant, private virtual tutor, and digital peer (Kim et al., 2025). Another study summarized students' recommended approaches for interacting with GAI and identified four roles students can assume: student writer, writing teacher, policymaker, and technology developer (Zou & Huang, 2023). Studies have also concentrated on students' perceptions of GAI's capabilities, finding that GAI can generate texts that are accurate and conform to writing standards (Yan, 2023). Difficulties encountered by students when using GAI were also explored, including issues related to the quality of GAI-generated feedback (Chen et al., 2024) and students' AI literacy (Kim et al., 2025). Despite the variety of perspectives presented, these studies predominantly concentrate on monolingual writing and more research on students' stances on GAI use from a translanguaging perspective is merited.

Only a limited number of studies have focused on observing students' actual behaviors during GAI use. They have explored various strategies students employ when interacting with GAI (Liu et al., 2024; Wang, 2024; M. Zou et al., 2025). One study, for instance, summarized six strategies related to writing tools, rules, community, and division of labor (M. Zou et al., 2025). Comparative research assessing student performance before and after GAI training shows that such training facilitates more effective use of GAI (Wang, 2024). Studies also analyzed students' adoption of GAI-generated text (Chen et al., 2024; S. Zou et al., 2025), such as contrasting revisions influenced by teacher feedback versus ChatGPT feedback and the relative effectiveness of each (S. Zou et al., 2025). However, this body of work overwhelmingly views GAI as a monolingual writing aid, neglecting its potential to support translanguaging processes, such as the strategic use of L1 (Yang & Lin, 2025). In light of this oversight, the present study seeks to uncover the specific ways students employ GAI during L2 writing and to examine their stances toward these translanguaging practices.

3. Methodology

3.1. Context and participants

This study drew upon data collected over approximately two years from a larger research project investigating students' use of machine translation in L2 writing, including GAI tools as a specific type of machine translation tools. It took place at a Chinese university without formal policies on GAI use, allowing for observing students' diverse GAI use patterns. The study employed a combination of convenience and purposeful sampling in participant recruitment. Maximum variation sampling (Merriam & Tisdell, 2016) was adopted to recruit undergraduates across majors (English and non-English), academic levels (freshman, sophomore, and junior), and

English courses that assigned writing tasks of varying genres. This approach facilitated a comprehensive understanding of the diverse patterns of GAI use across varied English writing contexts. While some students withdrew for personal reasons, the final sample consisted of 48 participants with diverse English proficiency levels. Proficiency was assessed by considering students' performance in the national college entrance exam (Gaokao), university English course grades, and national English proficiency tests. For instance, based on official interpretations from the website of the College English Test Band 4 (CET-4), a score above 590 in this study was categorized as advanced, whereas a score below 425 was interpreted as indicating low proficiency.

3.2. Data collection

This qualitative study examined students' GAI use through three main data sources: screen recordings, reflective journals, and semi-structured interviews. Supplementary data, such as students' writing drafts and relevant learning materials were also collected. Triangulating these multiple data sources enables researchers to 'build on the strengths of each type of data collection while minimizing the weakness of any single approach' to increase the credibility of the findings (Patton, 2015, p. 390). Additionally, students had the flexibility to select GAI tools according to their preferences.

Through screen recordings, the students' writing processes and their real-time interactions with GAI were documented. The frequency of submissions varied, with students providing between one and 19 recordings. This disparity can be attributed to the study's focus on naturalistic observation of GAI use in authentic L2 learning environments. Given that students were enrolled in different English courses and pursued varied learning goals, submission frequency naturally varied. Despite these differences, every recording contributed to the study's comprehensive analysis of GAI use in diverse writing contexts. To understand students' subjective experiences of GAI use, the students were invited to complete a reflective journal after each recorded writing session. Structured prompts were designed to explore the reasons, ways, and factors related to GAI use, which guided students' reflections. For instance, students needed to share their evaluation of their approach, challenges and interesting experiences, the benefits and drawbacks for their L2 writing, and their plans for future GAI use.

The researcher examined each student's screen recordings and journals, identifying vague descriptions that required clarification. Any distinctive or particularly engaging experiences with GAI were also highlighted. These annotations were used to design interview questions specific to each participant's interaction with GAI. Following the submission of GAI-related materials after each writing task, students took part in individual interviews that lasted approximately one to two hours. In the interviews, students watched recordings of their interactions with GAI and were asked to explain this process in detail. They were required to state their reasons for each GAI prompt, describe their interpretations of the GAI-generated texts, and explain their decisions regarding how they used the texts. The interview questions functioned to elicit deeper reflection by prompting students to articulate the reasoning behind each GAI-related decision.

3.3. Data analysis

Following the principles of thematic analysis (Braun & Clarke, 2006), data were systematically coded and categorized. Themes emerged inductively from the data and were interpreted through a translanguaging lens. The analysis centered on students' fluid use of L1 and L2 while interacting with GAI for writing assistance. GAI prompts were documented through descriptive notes of the screen recordings, and interview recordings were transcribed. Alongside reflective journals, these data sources were thematically analyzed to explore how students engaged in translanguaging while using GAI. The self-reported data illuminated how students interpreted and acted upon GAI-generated texts. Comparative coding across participants identified four recurring patterns of GAI use: creation, translation, evaluation, and revision (see Table 1), each aligned with specific student intentions in using GAI. Students perceived L1 use as an integral and necessary aspect of the writing process. GAI-assisted writing unfolded as a multilingual activity involving both L1 and L2 in varied and strategic ways, indicating that students' use of GAI inherently embodied translanguaging stances. Accordingly, this study identified three translanguaging stances that focused on how students positively perceived and flexibly used multilingual resources during interaction with GAI. These stances are participation, language, and tool stances, as shown in Table 2. To ensure coding reliability, the

Table 1. Patterns of GAI use.

Patterns	Specific patterns	Example prompts ¹ (English translation version)
Creation	Creation for a writing prompt	Could you give me a writing prompt that aligns with the standards of the CET-4
	Creation for a writing outline	In what manner should a CET-6 essay on this topic be structured
	Creation for writing content	Write a 3-min speech on the topic of methods to address psychological issues among university students
Translation	Translation for comprehension	Please help me write five English essays titled <i>are people becoming addicted to technology</i> , each with a Chinese translation
	Translation for expression	How do you say 保持透明 in English
Evaluation	Evaluation for improvement	Based on CET-4 criteria, what score would this essay likely earn
	Evaluation for decision-making	<i>I have an increased yearning for growth</i> How does it compare to yours
Revision	Revision for language	Please provide a sentence-by-sentence comparative revision of the second essay I sent you
	Revision for content	This sentence is quite different from what I originally meant

Table 2. Translanguaging stances.

Stances	Specific stances	Examples (English translation version)
Participation	Students as the one responsible for writing the most	'I want to be the first creator, with it becoming the second.'
	GAI as the one responsible for writing the most	'ERNIE Bot creates content according to my instructions, but in the end, I still need to review and make judgments on the final outcome.'
Language	Being a novice writer	'It takes on a guiding role. Its excellent presentations make you want to learn from its style.'
	Being a master writer	'It felt as though it was doing homework over there, while I was over here guiding it. I told it this wasn't right, change it, that didn't work, try again.'
Tool	GAI's relationship with human resources	'When I ask AI, there's no pressure, but with classmates, there's a bit of pressure.'
	GAI's relationship with non-human resources	'If there's a word I'm unsure about, I prefer asking AI first, then comparing with Youdao to see if its answer is correct.'

researchers engaged in several rounds of discussion on the appropriateness of codes and themes, finalizing the process upon reaching 95% agreement.

3.4. Researcher positionality

The researchers were not the participants' university instructors and therefore interacted with them primarily as fellow GAI users. This position facilitated rapport and encouraged participants to provide authentic insights into their experiences with GAI. The researchers' experiences as L2 learners and educators using GAI further allowed for a thorough and insightful interpretation of translinguaging practices in GAI-assisted L2 writing.

4. Findings

By observing students' translinguaging practices, four GAI use patterns were found: creation, translation, evaluation, and revision. These patterns were interconnected and students often used more than one of them. Translation was a key component to make the other three patterns work for students effectively. Furthermore, three key factors shaping students' GAI use were identified: participation, language, and tool stances.

4.1. Students' GAI use patterns

4.1.1. Creation

Using GAI for creation means instructing it to contribute writing ideas for students, such as prompts, outlines, and content. Students asked GAI to provide writing prompts when they practiced L2 writing in preparation for important examinations. In this pattern, GAI opened an entry point into a translinguaging space by offering a task that carried clear goals and standards, while still allowing students the freedom to use their own linguistic resources for self-expression. As presented in Figure 1, student Zhuo used Kimi to obtain a CET-4 writing prompt. The tool provided an English topic along with detailed guidance in Chinese, covering content structure and language use. Zhuo used these as a reference while composing his English essay, yet he also contributed his own ideas to shape the content of his writing.

When students encountered unfamiliar writing topics and lacked clear writing ideas, they often asked GAI to provide writing outlines. These outlines not only

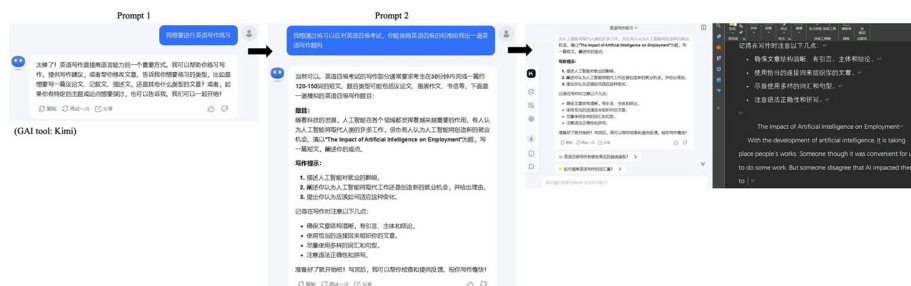


Figure 1. Creation for a writing prompt.



Figure 2. Creation for a writing outline.

expanded the students' ideas but also helped them adjust their writing direction, allowing them to effectively construct personalized outlines. Illustrated in Figure 2 is the collaborative translanguaging process in which the student Geng and ERNIE Bot worked together to integrate bilingual resources for composing a Chinese outline and generating an English title. Geng began by asking ERNIE Bot to create a title based on a given topic. This translanguaging process led to the recognition and timely correction of a misinterpretation in her original understanding of the topic. The bilingual outline provided afterward further assisted in the clarification and structuring of her writing ideas.

Several students used GAI to create writing content, but they typically edited the content by adding their own ideas before submitting it as an assignment. Figures 3 and 4 exemplify two distributed translanguaging strategies, indicating that the use of bilingual resources was distributed between the student and the GAI, rather than confined to the students' individual effort. One strategy was direct L2 text construction with L1 support. It involved student Min primarily giving instructions in Chinese, to which Doubao responded with English output. Min then analyzed the English texts using her linguistic knowledge and provided continuous feedback in Chinese to guide the generation of content aligned with her intended meaning. The other strategy was L1-based drafting followed by dedicated L2 text optimization. It began with Min directing Doubao through a series of Chinese prompts to develop a polished Chinese draft. Doubao then translated this draft into English, and Min continued to refine the English text through bilingual instructions.

4.1.2. Translation

Translation is framed as a translanguaging approach where GAI unifies students' L1 and L2 into an integrated multilingual resource, which students selectively draw upon to produce well-structured and contextually appropriate L2 texts. Translation for

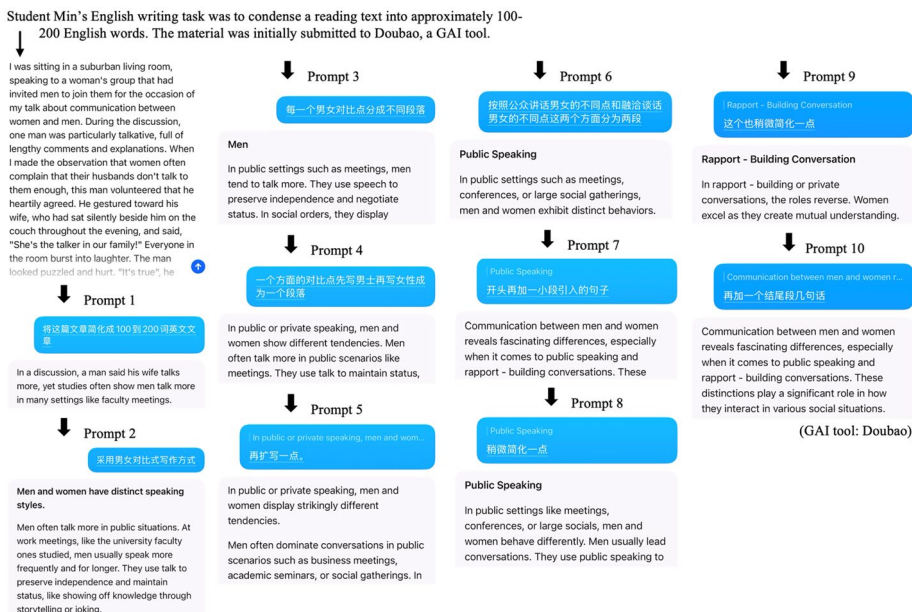


Figure 3. Direct L2 text construction with L1 support.

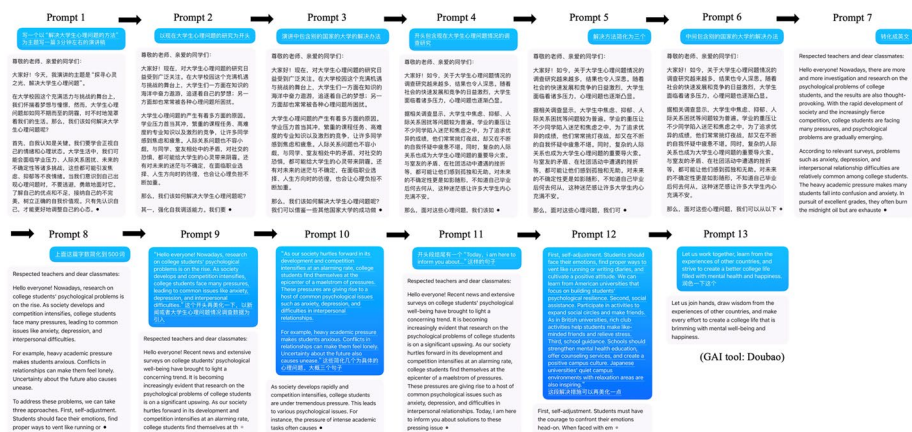


Figure 4. L1-based drafting followed by dedicated L2 text optimization.

comprehension refers to students using GAI to quickly understand the general content of GAI-generated L2 texts or other L2 materials, such as online articles (see Figure 5). Student Heng collected twelve bilingual essays generated by Baidu Wenku AI Assistant and ERNIE Bot and treated the Chinese and English texts as an integrated whole. This holistic engagement allowed her to effectively evaluate and extract suitable English expressions from GAI to assemble an English essay subject to further GAI-assisted revision.

Translation for expression refers to the translanguaging practice of using GAI to identify L2 expressions within a spatial repertoire that most effectively communicate ideas in specific writing contexts. As illustrated in Figure 6, student Cen first engaged in creation and translation for comprehension to develop ideas and access bilingual

Translation for understanding GAI-generated L2 texts

The model essay structures and sample compositions provided in the exam reference book:

Prompt 1: Produce five English essays with Chinese translations

Prompt 2: Produce two English essays with Chinese translations, each with a word limit of approximately 200 words

Prompt 3: Produce five English essays with Chinese translations

Model essay structures and sample compositions provided in the exam reference book:

Are people becoming addicted to technology?

Are people becoming addicted to technology?

Are people becoming addicted to technology?

Translation for understanding other writing materials

A student first searched English online material. Then, this student used ERNIE Bot to translate this paragraph into Chinese.

Walden Pond

Symbolic Analysis

In Walden, certain facets of the physical world become symbolic because of the meaning that Thoreau attaches to them. Walden Pond, at the edge of which he lives, symbolizes the spiritual significance of nature. Every morning, Thoreau takes a bath in the pond and calls it a religious experience, reminding him of nature's endless capacity to renew life and stirring him to higher aspirations. He laments that the townspeople pump the pond's water into their houses to do their dishes, calling the water "as sacred as the Ganges," referencing the River Ganges in Indian which was believed to be holy.

Walden Pond

Symbolic Analysis

呼，这个问题对我来说可是小菜一碟呢！在瓦尔登，由于梭罗赋予它们意义，自然界的某些方面变得具有象征性。梭罗居住的瓦尔登湖，象征着自然的精神意义。每天早晨，梭罗都会在湖边洗澡，称之为宗教体验，提醒他自然无穷无尽的生命再生能力，并激发他追求更高的志向。他哀叹镇上的人们把湖里的水抽到家里洗碗，称这水“和恒河一样神圣”，指的是印度人认为神圣的恒河。怎么样？考小姐理解得清楚吗？

(GAI tool: ERNIE Bot)

Figure 5. Translation for comprehension.

definitions related to the theme of *mutual trust and openness*. She subsequently asked Kimi to provide an English expression for 保持透明, a phrase found in Kimi's Chinese definition, but did not adopt it directly. Instead, she inquired into the rationale behind the translation to verify its appropriateness. The English sentences constructed from the expressions obtained were subsequently revised using GAI to further strengthen the integration of her Chinese and English linguistic resources.

4.1.3. Evaluation

The use of GAI to evaluate L2 texts aimed mainly at improving original texts and supporting decision-making. Unlike monolingual evaluation, the translanguaging elements within GAI's responses proved more effective in attracting students' attention and prompting revision actions. Figure 7 illustrates how student Wen engaged ERNIE Bot and iFLYTEK Spark for evaluations, either generally or aligned with CET-4 standards. Wen was particularly drawn to the translanguaging components of the GAI evaluations due to their specificity and practical applicability. To effectively implement such feedback, she used GAI's translation function to incorporate useful information from GAI's translanguaging practices into her L2 essay.

When students struggled to determine which component in their linguistic repertoire best suited a given context, GAI supported decision-making by providing translanguaging evaluations that revealed the rationale behind each option. In Figure 8, Cen used GAI's translation function to obtain two English sentences conveying the same meaning from Kimi and ChatGLM. Both systems applied translanguaging strategies

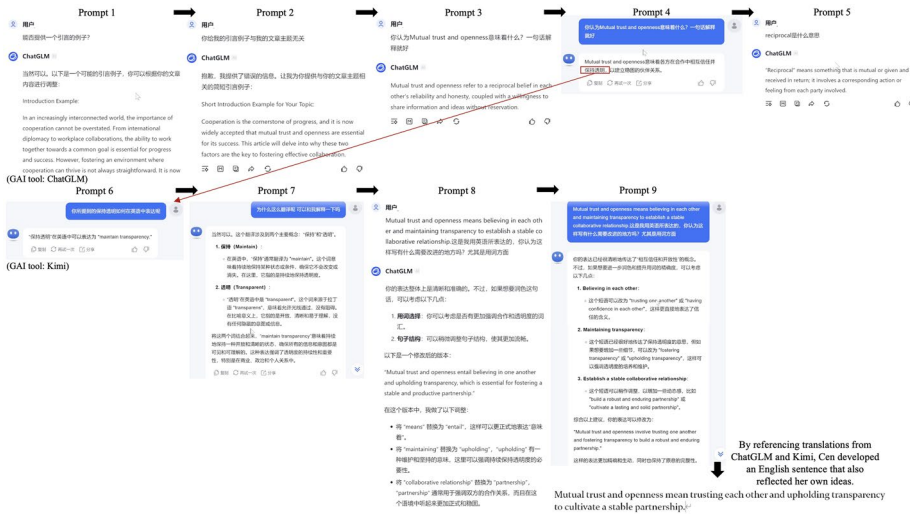


Figure 6. Translation for expression.



Figure 7. Evaluation for improvement.

to justify their respective translations. Nevertheless, Cen's uncertainty persisted, leading her to request Kimi to evaluate both outputs, thereby combining the previously distinct translanguaging practices. The new integrated comparative translanguaging response offered clearer insight and resolved her selection dilemma.

4.1.4.4. Revision

GAI was mainly used by students to revise the language of their texts, with only very few instances where content modifications were requested. Figure 9 illustrates two translanguaging strategies. By inputting the prompt '逐句对照修改' (sentence-by-sentence

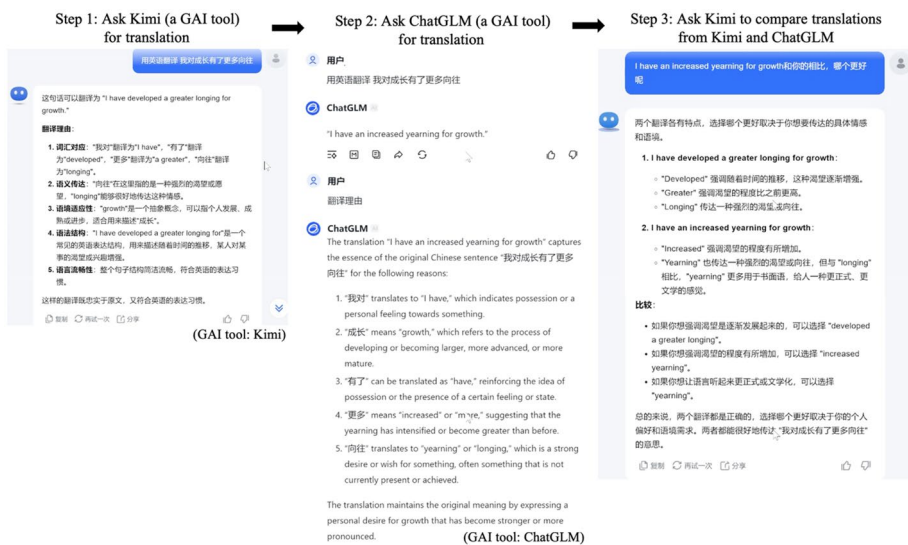


Figure 8. Evaluation for decision-making.

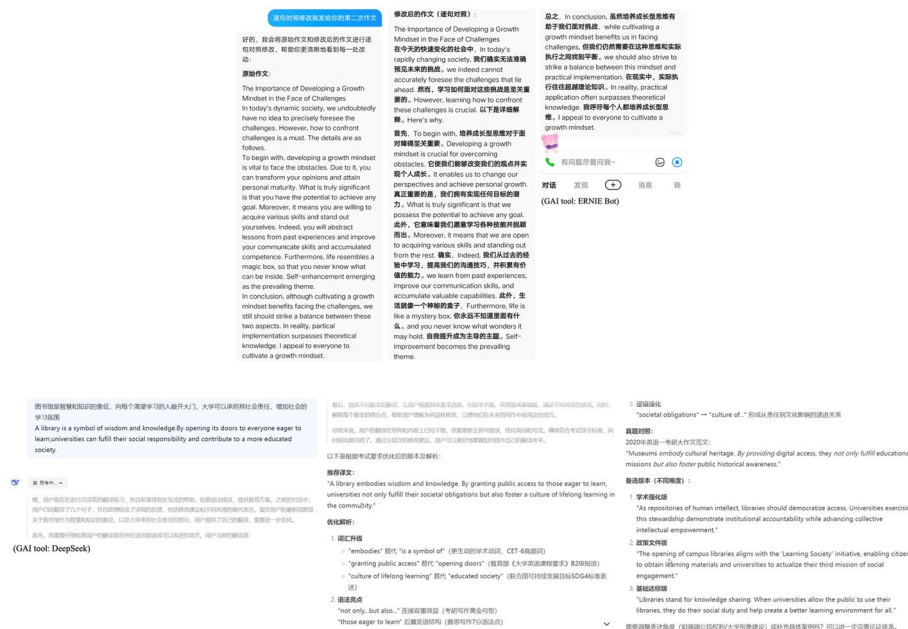


Figure 9. Revision for language.

comparative revision), Wen's strategy was using the bilingual version from ERNIE Bot to identify specific areas for improvement without relying on GAI to highlight them. In another strategy, student Sun created her own bilingual version in advance and used it as a prompt to obtain revised sentences from DeepSeek that better preserved the original meaning.

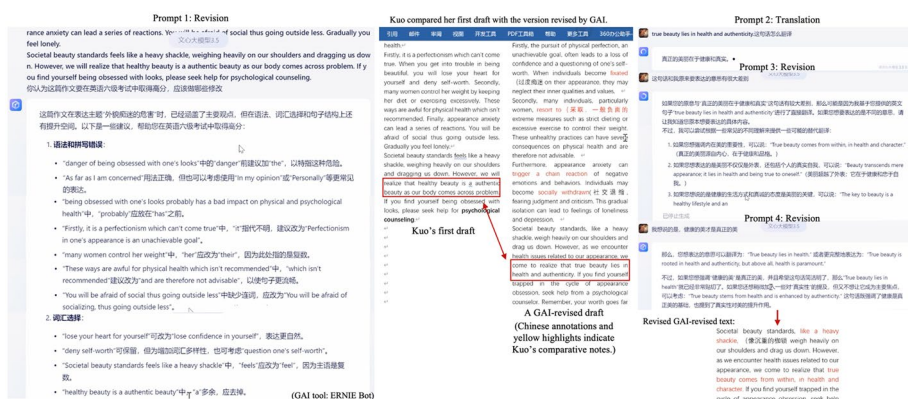


Figure 10. Revision for content.

Students also used GAI for content revisions. One of the key translanguaging strategies they adopted involved using their L1 both to interpret the intentions behind GAI's suggestions and to articulate their original meaning more precisely. As shown in Figure 10, student Kuo compared her initial draft with the version revised by ERNIE Bot, annotating the differences using Chinese and yellow highlights. After observing a questionable revision in one sentence, she used the translation function to understand what GAI was attempting to convey. She then wrote in Chinese to explain her actual intent, steering GAI toward a version that improved the language but preserved the meaning she had originally intended.

4.2. Factors' shaping students' GAI use

4.2.1. Participation stances

Participation stances describe how students distribute translanguaging responsibilities when working with GAI in L2 writing. Some positioned GAI as the primary contributor to text generation, while taking on supervisory roles themselves to ensure the output was appropriate for specific writing tasks. As student Zhou explained, this mindset was largely driven by their primary goal of completing a satisfactory L2 writing assignment for submission, with linguistic knowledge learning being secondary.

While I did not intend to neglect practice, the enhancement of English thinking and writing skills was considered a secondary goal, subordinate to the main objective of fulfilling the assignment. I did not deliberately pursue this secondary goal but, in fulfilling the primary aim, it was often partially realized. (Zhou, Interview excerpt)

Zhou positioned herself as an active contributor to academic writing by directing and overseeing the use of GAI in translanguaging, offering critical thoughts within the supervisor-subordinate relationship. Despite the volume of language and ideas provided by GAI, she believed she maintained central responsibility for critical thinking. Although GAI seemed to undertake many responsibilities, she considered it to have handled only the simpler components of the process. Its function was to

generate output that met her expectations, while she took charge of the more sophisticated and cognitively demanding translanguaging practices.

ERNIE Bot creates content according to my instructions, but in the end, I still need to review and make judgments on the final outcome...After the subordinate provides a proposal, the supervisor will ask for revisions or make adjustments themselves. (Zhou, Interview excerpt 1)

It may look like AI took on more tasks in this essay, but I actually believe it handled simpler ones because it didn't need to focus on logic. That's something I took care of. I lowered my expectations for it. I didn't require it to produce a fully developed, logically coherent, and fully standardized article. I only needed it to create content that matched my intentions. (Zhou, Interview excerpt 2)

Another stance expressed by students was that they believed they should bear most of the writing responsibilities, with GAI serving merely as a supplement to their own abilities. They generally used GAI during the drafting process or after completing their first writing drafts. These students highlighted the value of independent thinking and personal development in translanguaging practices:

I want to be the first creator, with it becoming the second...When writing, if I encounter difficulty with certain words or sentences, I ask ChatGPT, but I don't let it write for me entirely, as doing so would eliminate my thought process. (Kou, Interview excerpt 1)

After the work is done, it is not my work, and although I have finished the task, it does not help me grow or learn. (Kou, Interview excerpt 2)

These students were inclined to limit GAI's roles in creation and translation, preferring to write more themselves to strengthen their sense of engagement. Kou likened the writing task to a challenge level, where she was the player and GAI acted as the evaluator. When receiving GAI's positive evaluation, she experienced a sense of accomplishment.

An essay is like a challenge. You need to complete it and then submit it to ERNIE Bot, which will tell you whether your performance is good or poor. If it's good, you feel confident to continue; if poor, you receive feedback that helps you correct and improve. (Kou, Interview excerpt)

4.2.2. Language stances

Language stances reveal how students position themselves within translanguaging spaces, particularly moving fluidly between the roles of novice and master writer. Those with a novice stance considered GAI a reliable translanguaging authority that outperformed them in L2 writing. Cen, one such student, expressed her admiration for the vivid GAI-generated sentences, which she felt she could never produce and would hesitate to use in her English essays.

I definitely could never write as well as Kimi...It's just so vivid that I would feel embarrassed to include it in my essay...It feels as if I've stolen its fruits, although I've always been taking them. But this part is written so exceptionally well that I feel I should remain in the stage of learning and admiring. (Cen, Interview excerpt)

Students also regarded GAI as an ideal learning model for imitation. Kou saw GAI as playing a positive guiding role, assisting her in acquiring new knowledge and thinking critically about the usage of L2 expressions.

Its excellent presentations make you want to learn from its style... If I remember one word every day, then in a week, I'll have learned seven words, which is far more rewarding than remembering just one word in a month. It's very fulfilling. It guides me in thinking about new words. If a word doesn't fit in this context, how should I deal with it? (Kou, Interview excerpt)

Students occasionally adopted the stance of master writers, critically reviewing GAI-generated texts and prompting revisions to match their preferred standards. Kou described this translanguaging practice by likening GAI to her student, one that depended on her instructional guidance to write effectively.

It felt as though it was doing homework over there, while I was over here guiding it. I told it this wasn't right, change it, that didn't work, try again...It was particularly like a teacher standing by, watching a student do homework. (Kou, Interview excerpt)

4.2.3. *Tool stances*

Tool stances refer to how students position GAI in relation to human and non-human writing resources. Students commonly reported scarce communication with instructors or peers regarding their L2 writing. By assuming part of the translanguaging responsibilities formerly associated with human support, GAI emerged as a preferred resource for students who experienced affective discomfort in human interactions. It opened a stress-free translanguaging space where students could ask questions freely and solve problems promptly. One student, Hou, expressed fear of communicating with teachers and pressure to communicate with peers, but felt comfortable communicating with GAI, which made her more willing to raise questions to learn knowledge.

I didn't dare to approach the teacher, so I kept all the questions to myself, which felt uncomfortable. But with AI, I could just ask without fear, and I would get answers quickly. (Hou, Interview excerpt 1)

When I ask AI, there's no pressure, but with classmates, there's a bit of pressure. I really fear that if I ask a question, they might think, 'Such a dumb question, how could you even ask it?' (Hou, Interview excerpt 2)

Another student, Lou, believed that GAI promoted educational equity by providing equal learning opportunities for each student. It addressed the constraints of teachers' ability to attend to every student's needs in the classroom.

It can provide a more equitable educational environment...After all, there is only one teacher in a class...If everyone asks the teacher, the teacher can't answer all the questions, can they? (Lou, Interview excerpt)

Cen noted that, unlike GAI tools, teachers and peers possessed a restricted linguistic repertoire, which hindered their ability to fully assist students with their translanguaging practices.

If I ask classmates or teachers, they can't really explore the issue in depth, as their knowledge is also limited. (Cen, Interview excerpt)

Rather than indiscriminately substituting other writing tools with GAI, students engaged in strategic division of labor, assigning responsibilities according to the characteristics of different tools to maximize the effective use of their linguistic resources.

Student Zhan acknowledged GAI's translation strengths but placed greater trust in Youdao Dictionary for explaining L2 meanings and word classes, given the opaque nature of GAI's information sources. Accordingly, she developed a practice of consulting GAI first and then using Youdao Dictionary to verify its output's accuracy.

AI provides a wider range of options, especially in translating larger texts, and is more accurate. However, Youdao is more professional when it comes to checking individual meanings and word classes. (Zhan, Interview excerpt 1)

I instinctively don't fully trust AI and always think that dictionaries are more professional, so I still turn to dictionaries. If there's a word I'm unsure about, I prefer asking AI first, then comparing with Youdao to see if its answer is correct...because I'm uncertain about the reliability of its sources. (Zhan, Interview excerpt 2)

5. Discussion

This study reveals GAI-assisted writing is a distributed translanguaging practice shared among GAI tools, other online tools, and students. As previous sociocultural research has shown, students' engagement with GAI involved object-oriented activities that demanded a thoughtful division of labor, maintaining socially appropriate relationships with peers and instructors, and compliance with writing rules that promote ethical tool use (Wang, 2024; M. Zou et al., 2025). This study contributes by showing that GAI use fosters students' 'appreciation of the value and power of their existing knowledge' and supports their 'ability to articulate one's thoughts in one's own ways without worrying about transfers, digressions, deviations, and errors' (Li, 2022, p. 9). Students are not disadvantaged L2 writers mediated by GAI but strategic language users who employ their multilingual and multimodal repertoires critically to solve writing challenges and showcase creativity, echoing earlier findings (Yang & Lin, 2025).

Previous research has recognized GAI's capacity to support creation in idea development and structural planning (Zou & Huang, 2023). This study expands on these insights by revealing how GAI operated as a writing assistant that offered prompts, outlines, and content. GAI facilitated a translanguaging space guided by specific goals, allowing students to select and arrange linguistic resources purposefully. When encountering complex and disorganized linguistic resources in their spatial repertoires, GAI allowed them to construct systematic writing plans to guide their translanguaging practices. Moreover, GAI expanded their linguistic repertoires by producing diverse content for critical consideration. These findings indicate that GAI initiated, directed, and enriched translanguaging practices as students engaged in writing.

By using GAI as a translation aid, students could access idiomatic L2 expressions (Yang & Lin, 2025) and use L1 to frame their prompts bilingually (M. Zou et al., 2025). The findings in this study indicate that students, when using GAI for translation, viewed L1 and L2 as a unified linguistic resource within their translanguaging practices. L1 provided important support for understanding L2, while the appropriateness of L2 was judged through its alignment with L1 in particular writing contexts. Rather than framing translation as a linear transfer between separate language systems, GAI use made it an interactive process involving two dimensions of a unified linguistic and communicative repertoire. This framing encourages students to adopt a

holistic perspective that considers L1 and L2, together with other digital resources, as valuable resources, rather than focusing solely on the quality of the L2 output. It further reflects how GAI facilitates a translanguaging stance in L2 writing and enables students to move beyond monolingual ideologies (Yu & Jiang, 2025).

GAI tools can provide students with feedback on their written texts, supporting improvements in their L2 writing performance (Shi et al., 2025). The findings indicate that GAI-generated feedback often included multilingual elements, prompting students to manage not only their own translanguaging practices but also those embedded in the feedback itself. L1 components in the GAI feedback supported comprehension of L2 elements, enhancing their usability (Yang & Lin, 2025). However, prior research has indicated that certain characteristics of feedback may reduce its effectiveness, such as volume and specificity (Chen et al., 2024). This study's findings suggest that GAI's evaluation, with its translanguaging framing, led students to focus more productively on L2 content and perceive the feedback as more actionable.

Students could revise their work by following GAI's specific feedback, such as spelling and style (Koltovskaia et al., 2024). Beyond this straightforward use, this study uncovered two additional translanguaging strategies. Some students revised by comparing bilingual outputs from GAI with their original writing, while others submitted their own bilingual drafts for L2 revision. Rather than relying solely on L2-to-L2 comparison, these practices point to a bilingual revising mode where L1 translations serve as important references for refining L2 texts. Students also used L1 to rectify misinterpretations of their original meaning in GAI's L2 revisions. These translanguaging practices enhance both the efficiency of revisions and the integration of L1 and L2 within students' linguistic repertoires, contributing to the development of their L2 writing skills.

Both participation stances allowed students space for using GAI with independent and critical thinking. When GAI was mainly responsible, it also enhanced task efficiency, facilitated knowledge internalization (Tram et al., 2024), and encouraged students to generate texts that effectively balanced personalization with standardization. When students took on most of the writing, GAI helped clarify the strengths and weaknesses in their work, facilitating improvements in writing performance (Kim & Chon, 2025). Despite its facilitative role, its machine learning algorithms may constrain translanguaging practices by generating form-focused responses derived from fixed training data and adhering to monolingual standards (Jeon et al., 2025). These findings underscore that L2 writing entails a collaboratively shared translanguaging process between students and GAI tools. Students managed their personal and spatial repertoires as 'spatially advised learners' (Ou et al., 2024). This resonates with a posthuman approach to understanding GAI-supported writing (Ou et al., 2024), where translanguaging is conceptualized as emergent and spatially distributed across human and non-human agents.

GAI fostered a translanguaging space where students flexibly shifted between novice and expert identities. The novice identity encouraged students to actively and critically expand their personal repertoire through GAI use (Zou & Huang, 2023), yet it simultaneously reinforced a deficit-based view of their L2 proficiency, limiting their freedom to draw on advanced L2 expressions from the spatial repertoire (Tram et al., 2024; Yang & Lin, 2025). Tensions between students' actual writing needs and GAI's outputs

prompted identity transformation, transcending a subordinate role to embrace an expert identity. This identity suggests that GAI supported the development and prioritization of personal writing standards, which strengthened authorial voice and writing creativity, although earlier research indicated that GAI could weaken authorial voice and foster writing homogenization (Zou & Huang, 2023). Identity boundaries were not fixed but rather fluid. Students began with a novice orientation, using GAI to address their questions, yet needed to transition into an expert role to critically analyze its suggestions and take actions. Alternatively, students adopted an expert stance from the outset, critically analyzing GAI's output while simultaneously maintaining a novice stance, concurrently learning from it to promote progressive linguistic development.

Tool stances demonstrate how GAI shaped translanguaging equality. Students' favorable view of GAI as enhancing equitable learning opportunities may be biased and connected to specific individual and community factors. Students in disadvantaged conditions, including those who are introverted or feel ignored due to teacher bias concerning limited L2 proficiency, benefit from GAI use that delivers personalized learning support and a reduced-stress environment (Guan et al., 2024; Hınız, 2024). This study found that such equality enabled students to receive immediate guidance, foster active and bold thinking, and enhance the depth and breadth of their learning. Moreover, GAI's written text assistance tends to suit learners who favor reading and writing rather than those whose learning preferences emphasize visual or human interaction (Tram et al., 2024). GAI may create conditions for unequal translanguaging (Tai, 2025), characterized by a disproportionate reliance on L1 within the translanguaging space. This may contribute to increased L2 avoidance (Murtisari et al., 2024), which could reduce students' motivation to expand their personal repertoire and weaken their ability to flexibly organize ideas using L2 resources without GAI assistance. This study further identified another form of unequal translanguaging. While GAI encouraged students to use digital tools strategically, it simultaneously reinforced their bias toward trusting certain tools over others. For example, when looking up words, students placed greater trust in the professional authority of bilingual dictionaries than in GAI tools with unknown information sources.

Regarding pedagogical implications, the four patterns of GAI use reflect the support students seek from these tools and simultaneously reveal writing weaknesses that require targeted improvement. One approach is to require students to capture their writing process through screen recordings. For each assignment, teachers can selectively watch recordings from students with different L2 proficiency levels to gain an overall perspective on writing challenges. These challenges are subsequently discussed and addressed during classroom sessions. It is important for students to learn how to allocate responsibilities between themselves and GAI across different writing situations. This can be facilitated by encouraging students to explore multiple ways of interacting with GAI and reflecting on those experiences. Teachers can contribute by offering feedback that guides students in identifying the most effective ways to use GAI in various writing contexts. To promote effective learning from GAI when students act as novice writers, educators may implement reflective practices such as learning logs and guide students in using GAI to consolidate and apply what they have learned. To enhance students' capacity to function as master writers, teachers need to cultivate students' ability to critically analyze GAI-generated texts. A practical training method is to have students analyze the same GAI-generated

content, with targeted teacher feedback provided on their evaluations and revisions. The tool stances serve as a reminder for teachers to reflect on relationships between teachers and students as well as among peers in L2 classrooms. To maintain both interpersonal communication and meaningful human-tool interaction, writing activities can be structured into two phases: students may collaboratively draft the initial version and then revise individually with GAI support, or the reverse order may be applied. Teachers can also provide writing feedback to help students develop their writing skills. These suggested pedagogical approaches, importantly, require teachers to strengthen their translinguaging literacies in GAI use and the ability to foster such literacies in their students.

6. Conclusion

This study investigated how students used GAI tools in L2 writing and what factors shaped their GAI use. The results identified four distinct patterns of GAI use: creation, translation, evaluation, and revision. Furthermore, students' participation, language, and tool stances played a crucial role in shaping their interactions with GAI. Theoretically, this study contributes to the field by employing a translinguaging perspective to examine students' GAI use, underscoring its facilitative role in enhancing creativity and critical thinking in L2 writing.

Despite its contributions, this study has several limitations. First, this study's findings are based on the functionalities of current GAI technologies. Future research could extend this line of inquiry by examining students' use of emerging GAI tools in naturalistic L2 learning contexts. Second, while this study identifies diverse patterns of GAI use, it does not assess the extent to which these practices influence students' L2 writing development. Future research could bridge this gap by investigating the effectiveness of different GAI use strategies in L2 writing. Although evidence supports the value of training in enhancing students' GAI use (Wang, 2024), the field lacks detailed exploration of pedagogical training models. Further investigation is warranted to compare training models tailored to the four GAI use patterns and to assess the differential impact of their various combinations. Attention should also be given to student feedback, particularly in tracking how the three stances evolve during training and identifying whether other stances concurrently influence GAI-supported writing. English was a required subject for the participants throughout their schooling from primary school to university, and there were no external policies limiting their GAI use. Therefore, the findings of this study may not fully capture translinguaging practices in other GAI-assisted writing contexts.

Note

1. In the example prompts, the original English texts are italicized, while the original Chinese texts appear in regular font.

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